Kernels and Functions of Positive Type

Let X be a topological space

Definition: A kernel of positive type on X is a continuous function $\Phi : X \times X \to \mathbb{C}$ such that for any $n \in \mathbb{N}$, any elements $x_1, \ldots, x_n \in X$, and any $c_1, \ldots, c_n \in \mathbb{C}$,

$$\sum_{i=1}^{n} \sum_{j=1}^{n} c_i \overline{c_j} \Phi\left(x_i, x_j\right) \geq 0.$$

Equivalently, the matrix $\left(\Phi\left(x_i, x_j\right)\right)_{i,j} \in \mathbb{M}_n$ is positive.

Proposition: If Φ is a kernel of positive type on X, then for all $x, y \in X$, one has

(i) $\Phi(y,x) = \overline{\Phi(x,y)}$;

(ii) $\left|\Phi(x,y)\right|^2 \leq \Phi(x,x)\Phi(y,y)$.

Proof. When $n = 2$, $x_1 = x$, and $x_2 = y$, one observes that the matrix

$$A = \begin{pmatrix} \Phi(x,x) & \Phi(x,y) \\ \Phi(y,x) & \Phi(y,y) \end{pmatrix}$$

is positive, so it is hermitian with positive diagonal values satisfying $\Phi(x,x)\Phi(y,y) - \Phi(y,x)\Phi(x,y) \geq 0$. □

The canonical example is the following. If $f : X \to \mathscr{H}$ is a continuous map into a Hilbert space, then

$$\Phi(x,y) = \langle f(x), f(y) \rangle$$

is a kernel of positive type on X.

Theorem (GNS for Kernels of Positive Type)**:** Let Φ be a kernel of positive type on a topological space X. Then, there exists a Hilbert space $\mathscr{H}$ and a continuous map $f : X \to \mathscr{H}$ such that

(i) $\Phi(x,y) = \langle f(x), f(y) \rangle$;

(ii) $\overline{\text{span}}\{f(x) : x \in X\} = \mathscr{H}$.

This Hilbert space is unique up to isomorphism.

Proof. For every $x \in X$, we let $\Phi_x(y) = \Phi(x,y)$. These are each continuous functions. Let $V = \text{span}\{\Phi_x : x \in X\} \subseteq C(X)$. Given two elements of V

$$\varphi = \sum_{i=1}^{m} a_i \Phi_{x_i}$$
$$\psi = \sum_{j=1}^{n} b_j \Phi_{x_j},$$

we write

$$\begin{aligned} \langle \varphi, \psi \rangle &= \sum_{i=1}^{m} \sum_{j=1}^{n} a_i \overline{b_j} \Phi\left(x_i, x_j\right) \\ &= \sum_{j=1}^{n} \overline{b_j} \varphi\left(x_j\right) \\ &= \sum_{i=1}^{m} a_i \overline{\psi(x_i)}. \end{aligned}$$

The form $\langle \varphi, \psi \rangle$ is a positive sesquilinear form on V. We claim it is definite. Notice that $\langle \varphi, \Phi_x \rangle = \varphi(x)$,

and by Cauchy–Schwarz,

$$\left|\varphi(x)\right|^2 \leq \Phi(x,x)\langle\varphi,\varphi\rangle.$$

Completing V with respect to the induced norm yields the Hilbert space $(\mathscr{H},\langle\cdot,\cdot\rangle)$.

If $f: X \to V$ takes $x \mapsto \Phi_x$, then $\langle f(x), f(y)\rangle = \Phi(x,y)$, and f is continuous since

$$\left\|f(x)-f(y)\right\|^2 = \Phi(x,x)+\Phi(y,y)-2\,\mathrm{Re}(\Phi(x,y))$$

and Φ is continuous.

Now, suppose $(\mathscr{K},g)$ is another pair satisfying (i) and (ii). Then,

$$\begin{aligned}\left\|\sum_{i=1}^n \Phi_{x_i}\right\|^2 &= \sum_{i=1}^n\sum_{j=1}^n a_i\overline{a_j}\Phi\left(x_i,x_j\right)\\ &= \sum_{i=1}^n\sum_{j=1}^n a_i\overline{a_j}\left\langle g(x_i), g\left(x_j\right)\right\rangle\\ &= \left\|\sum_{i=1}^n a_i g(x_i)\right\|^2.\end{aligned}$$

Thus, if we define the map $V \to \mathscr{K}$ by taking $\Phi_{x_i} \mapsto g(x_i)$ and extending linearly, we obtain an isometry $T: \mathscr{H} \to \mathscr{K}$. Since $T(V) = \mathrm{span}\{g(x) : x \in X\}$, which is dense in $\mathscr{K}$, we have that $T \circ f = g$. □

We note that, given any kernels of positive type Φ and Ψ, the following are also kernels of positive type:

(i) $\overline{\Phi}$;

(ii) $t\Phi$ for all $t \geq 0$;

(iii) $\Phi + \Psi$;

(iv) $\Phi\Psi$;

(v) the pointwise limit of a family of kernels of positive type, given that the limit is continuous.

Thus, the kernels of positive type are a convex cone in the vector space of continuous kernels on X that is closed under conjugation, pointwise product, and pointwise convergence.

Lemma: Let Φ be a kernel of positive type on X such that $\left|\Phi(x,y)\right| < 1$ for all $x,y \in X$. Then, $(1-\Phi)^{-t}$ is a kernel of positive type for all $t \geq 0$.

Proof. The series

$$1 + \sum_{j=1}^{\infty} \frac{\prod_{k=0}^{j-1}(t+k)}{j!}\Phi(x,y)^j$$

converges to $(1-\Phi(x,y))^{-t}$ for all $x,y \in X$ by the assumption that $\left|\Phi(x,y)\right| < 1$ for all $x,y \in X$. □

A sort of dual to the kernels of positive type is a kernel of conditionally negative type. They are related to each other by Schoenberg's Theorem.

Definition: A real-valued kernel $\Psi: X \times X \to \mathbb{R}$ is said to be of *conditionally negative type* if

(i) $\Psi(x,x) = 0$ for all $x \in X$;

(ii) $\Psi(x,y) = \Psi(y,x)$ for all $x,y \in X$;

(iii) for all elements $x_1, \dots, x_n$ and all real numbers $c_1, \dots, c_n$ with $\sum c_i = 0$, one has

$$\sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j \Psi\left(x_i, x_j\right) \leq 0.$$

The canonical example is that of the map on a real Hilbert space

$$\Psi : \mathscr{H} \times \mathscr{H} \to \mathbb{R}$$
$$(\xi, \eta) \mapsto \left\|\xi - \eta\right\|^2.$$

One observes that for any $\xi_1, \dots, \xi_n \in \mathscr{H}$ and $c_1, \dots, c_n \in \mathbb{R}$ with $\sum c_i = 0$, we have

$$\begin{aligned}\sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j \Psi\left(\xi_i, \xi_j\right) &= \sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j \left\|\xi_i - \xi_j\right\|^2 \\ &= -2\left\|\sum_{i=1}^{n} c_i \xi_i\right\|^2.\end{aligned}$$

So, if X is a topological space and $\mathscr{H}$ is a real Hilbert space, with $f : X \to \mathscr{H}$ a continuous mapping, then the kernel $(x, y) \mapsto \left\|f(x) - f(y)\right\|^2$ is of conditionally negative type. Similar to how kernels of positive type arise, this is universal via a GNS construction.

Theorem (GNS for Kernels of Conditionally Negative Type)**:** Let Ψ be a kernel of conditionally negative type on X, and let $x_0 \in X$. There exists a real Hilbert space $\mathscr{H}$ and a continuous map $f : X \to \mathscr{H}$ such that

(i) $\Psi(x, y) = \left\|f(x) - f(y)\right\|^2$;

(ii) $\overline{\operatorname{span}}\{f(x) - f(x_0) : x \in X\} = \mathscr{H}$.

Moreover, this pair is unique up to affine isometry.

Proof. Let $V = \ker(\epsilon)$, where $\epsilon : \mathbb{R}[X] \to \mathbb{R}$ is the augmentation map

$$\sum_{i=1}^{m} c_i \delta_{x_i} \mapsto \sum_{i=1}^{m} c_i.$$

Let $\varphi = \sum_{i=1}^{m} a_i \delta_{x_i}$ and $\psi = \sum_{j=1}^{n} b_j \delta_{x_j}$ in V. Then, we let

$$\langle \varphi, \psi \rangle = -\frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} a_i b_j \Psi\left(x_i, x_j\right).$$

This defines a positive symmetric form on V. Set

$$N_\Psi = \{\varphi \in V : \langle \varphi, \varphi \rangle = 0\}.$$

By the Cauchy–Schwarz inequality, this is a linear subspace of V. Define

$$\langle [\varphi], [\psi] \rangle = \langle \varphi, \psi \rangle$$

on V/N_Ψ. Let $(\mathscr{H}, \langle \cdot, \cdot \rangle)$ be the Hilbert space completion of V/N_Ψ. Set $f : X \to H$ taking $x \mapsto \left[\delta_x - \delta_{x_0}\right]$. Then,

$$\left\|f(x) - f(y)\right\|^2 = \{si(x, y),$$

since $\Psi(x, x) = 0$ and $\Psi(y, y) = 0$ with $\Psi(y, x) = \Psi(x, y)$. Moreover, f is continuous since Ψ is continuous, so we obtain existence.

Let $(\mathscr{K}, g)$ be another such pair. We claim that the map

$$\left[\sum_{i=1}^{n} a_i \delta_{x_i}\right] \mapsto \sum_{i=1}^{n} a_i g(x_i)$$

extends to an affine isometry. Since $\sum_{i=1}^n a_i = 0$, it follows that

$$\begin{aligned}
\left\|\left[\sum_{i=1}^n a_i \delta_{x_i}\right]\right\|^2 &= \left\|\left[\sum_{i=1}^n a_i\left(\delta_{x_i} - \delta_{x_i}\right)\right]\right\|^2 \\
&= -\frac{1}{2}\sum_{i,j=1}^n a_i a_i\left(\Psi\left(x_i, x_j\right) - \Psi(x_i, x_0) - \Psi\left(x_j, x_0\right)\right) \\
&= -\frac{1}{2}\sum_{i,j=1}^n a_i a_j\left(\left\|g(x_i) - g\left(x_j\right)\right\|^2 - \|g(x_i) - g(x_0)\|^2 - \left\|g\left(x_j\right) - g(x_0)\right\|^2\right) \\
&= \sum_{i,j=1}^n a_i a_j\left\langle g(x_i) - g(x_0), g\left(x_j\right) - g(x_0)\right\rangle \\
&= \left\|\sum_{i=1}^n a_i (g(x_i) - g(x_0))\right\|^2 \\
&= \left\|\sum_{i=1}^n a_i g(x_i)\right\|^2.
\end{aligned}$$

Moreover, since $\operatorname{span}\{g(x) - g(x_0) : x \in X\}$ is dense in $\mathscr{K}$, it follows that T is surjective with $T \circ f = g$. □

Analogously to the case of kernels of positive type, there are certain inheritance properties for kernels of conditionally negative type (i.e., kernels of conditionally negative type form a convex cone and are closed in the topology of pointwise convergence on the space of continuous functions). Furthermore, if Φ is a real valued kernel of positive type, then $\Phi(x,x) - \Phi(x,y)$ is a kernel of conditionally negative type.

Lemma: Let X be a topological space, and let $\Psi : X \times X \to \mathbb{R}$ be a continuous kernel with $\Psi(x,x) = 0$ and $\Psi(y,x) = \Psi(x,y)$ for all $x, y \in X$. For a fixed $x_0 \in X$, let $\Phi : X \times X \to \mathbb{R}$ be defined by

$$\Phi(x,y) = \Psi(x,x_0) + \Psi(y,x_0) - \Psi(x,y).$$

Then, Ψ is a kernel of conditionally negative type if and only if Φ is a kernel of positive type.

Proof. Suppose Φ is a kernel of positive type. Then, for any $x_1, \ldots, x_n \in X$ and $c_1, \ldots, c_n \in \mathbb{R}$ with $\sum_{i=1}^n c_i = 0$, we have

$$\begin{aligned}
0 &\leq \sum_{i=1}^n \sum_{j=1}^n c_i c_j \Phi\left(x_i, x_j\right) \\
&= -\sum_{i=1}^n \sum_{j=1}^n c_i c_j \Psi\left(x_i, x_j\right),
\end{aligned}$$

showing that Ψ is of conditionally negative type.

Suppose Ψ is of conditionally negative type. Then, by the GNS for kernels of conditionally negative type, there is a real Hilbert space $\mathscr{H}$ and a continuous map $f : X \to \mathscr{H}$ such that $\Psi(x,y) = \left\|f(x) - f(y)\right\|^2$ for all $x, y \in X$. □

References

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